

2023

Time : 2 hours

Full Marks : 40

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer from **all** the section as directed.*

Section-A

Multiple choices questions

(Compulsory)

1. Choose the correct answer from the following
multiple choice questions : $1 \times 10 = 10$

(i) In the tossing three coins at a time the probability of getting at most one head is :

a) $\frac{3}{8}$

b) $\frac{7}{8}$

c) $\frac{1}{2}$

d) $\frac{1}{8}$

(ii) From a pack of 52 cards, two cards are drawn at random, the probability that one is an ace and other is a king is :

a) $\frac{1}{13}$

b) $\frac{1}{69}$

c) $\frac{16}{169}$

d) $\frac{8}{663}$

(iii) Probability can take value :

a) $-\infty$ to ∞

b) $-\infty$ to 1

- c) -1 to 1
 - d) 0 to 1
- (iv) Mean is a measure of :
- a) Location
 - b) Dispersion
 - c) Correlation
 - d) None of the above
- (v) If a constant value S_0 is subtracted from each observation of a set, the mean of set is :
- a) Increased by S_0
 - b) Decrease by S_0
 - c) Is not effected
 - d) Zero
- (vi) The correct relationship between AM, GM and HM is :
- a) $AM=GM=HM$
 - b) $GM \geq AM \geq HM$

c) $HM \geq GM \geq AM$

d) $AM \geq GM \geq HM$

(vii) Extreme value have no effect on :

a) Average

b) Median

c) Geometric mean

d) Harmonic mean

(viii) The term regression was introduced by :

a) R A Fisher

b) Sir Francis Galton

c) Karl Pearson

d) None of the above

(ix) In a regression line Y on X, the variable X is known as :

a) Independent

b) Regressor

- c) Explanatory variable
 - d) All of the above
- (x) The value of correlation ratio varies from :
- a) -1 to 1
 - b) -1 to 0
 - c) 0 to 1
 - d) 0 to ∞

Section-B

Short answer type questions

Answer any **four** questions of the following :

4×4=16

2. State addition and multiplication theorem of Probability.
3. Discuss Statistical and Mathematical definition of probability.

4. What do you understand by measure of central tendency ?
5. Define geometric mean with propertise.
6. How will you calculate median in case of ungrouped data ?
7. Define correlation coefficient.
8. Define regression coefficient.

Section-C

Long answer type questions

Answer any **two** questions of the following :

$7 \times 2 = 14$

9. The probability that an entering college student will be graduate is 0.4. Determine the probability that out of 5 entering students.
 - (a) None
 - (b) One
 - (c) At least one will be selected

10. Explain Bayes theorem with an example.
11. Define regression coefficient and how it differ from correlation.
12. Calculate the mean from the following table :

Class interval :	30-39	40-49	50-59	60-69	70-79	80-89
Frequency :	5	22	63	74	30	6

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